Hiten Dalmia

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Education

Chennai Mathematical Institute — Chennai, India
Master of Science in Theoretical Computer Science (On Full Scholarship) — *Aug. 2026 – Discontinued*

Kirori Mal College, University of Delhi — Delhi, India
Bachelor of Science (Honours with Research) in Mathematics, Minor in Economics — *Nov. 2022 – May 2026*

Experience

Full-Stack Developer — Mar. 2023 – Present
Freelance & Fractal — *Remote*

- Delivered 10+ production web and mobile applications for client businesses via Upwork and Fractal, specializing in Flutter cross-platform architecture, RESTful APIs, and payment gateway integration.
- **Order Procurement & Management Solution** | *Flutter, NextJS, ExpressJS, PostgreSQL, Stripe*
  - Built a full-stack order management portal using NextJS, ExpressJS, and PostgreSQL with session authentication, Dio HTTP handlers, and Stripe payment integration.
  - Designed a dual-tier interface with a customer ordering portal for live product listings and an administrative suite for task dispatch, audit logging, and recurring orders.
  - Automated order ingestion by parsing incoming WhatsApp client messages into structured PostgreSQL records via regex rules, eliminating manual entry and increasing processing speed.
  - Structured relational PostgreSQL schemas with foreign key constraints and B-tree indexes to optimize transaction throughput across customer and administrative order tables.

Flutter Developer — Oct. 2023 – Apr. 2025
Tagglabs Experimental Pvt. Ltd. (Contract) — *Gurugram, Haryana (Remote)*

- **Cross-platform AR App** | *Flutter, Dart, BLoC, Clean Architecture, OpenCV, Django* **(Live link)**
  - Architected a native C/C++ bridge using `dart:ffi` to port OpenCV SIFT algorithms into Flutter, structuring codebase via Clean Architecture for iOS and Android (Kotlin) platforms.
  - Engineered a real-time camera feed pipeline normalizing OS frame buffers and aspect ratios, converting video streams to C structs with zero memory leaks at 30+ FPS.
  - Implemented BLoC state management to stream match coordinates asynchronously from C algorithms to Dart, driving dynamic AR UI overlay rendering with sub-50ms latency.
  - Constructed responsive Flutter interfaces from Figma specs and connected 15+ REST APIs in Django using Dio, delivering location-aware spatial discovery features.
- **Kia National Conference App** | *Flutter, Dart, Firebase, BLoC*
  - Deployed a mobile app in Flutter for 500+ attendees at the Kia Bangkok Conference, integrating Firebase Push Notifications, BLoC state management, and Mockito unit test coverage.

Technical Skills

Mobile & Frameworks: Flutter, Dart, BLoC, Provider, GetX, Clean Architecture, MVVM, React Native, NextJS, Django
Backend & Testing: RESTful APIs, Dio, Firebase (Auth, Firestore, Crashlytics), Mockito, PostgreSQL, NodeJS, ExpressJS
Languages & Vision: Dart, Python, JavaScript/TypeScript, C/C++, Kotlin, Swift, SQL, OpenCV (SIFT)
Developer Tools & Cloud: Git, GitHub, JIRA, Figma, Google Play Store, App Store, Docker, Linux, AWS, LaTeX

Achievements

Erasmus Mundus Joint Master MATHS-DISC Fellowship — Europe
Issued by European Union and affiliated universities — *Mar. 2026*

Master in Mathematics (ISI) and MSc Computer Science (CMI) Scholarships — India
Indian Statistical Institute and Chennai Mathematical Institute respectively — *Jun. 2026*

Publications

Computable Analysis: Mathematical Foundations and Computability of Real Functions
Poincare Journal of Mathematical Physics & Applications, Vol. 1, No. 1 (2026), 1–20 (with Sujit Kumar)